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NO.1 PVT. UNIVERSITY IN ACADEMIC REPUTATION IN INDIA

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PERFECT SCORE OF 150/150 AS A TESTAMENT TO EXECEPTIONAL E-LEARNING METHODS

UPES UNIVERSITY OF TOMORROW

Elements of AIML

Unit 04 – Lecture 01: Supervised Learning (Regression)

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Topic focus: predicting **continuous values** and evaluating regression models

Repository: `https://github.com/tali7c/Elements-of-AIML`

Sources: Mueller and Massaron, *Machine Learning for Dummies*, For Dummies, 2016.;James, Witten, Hastie, and Tibshirani, *An Introduction to Statistical Learning* (2e), selected chapters.;Scikit-learn documentation, accessed 2026-02-28.

Quick Links

Regression Linear Model Metrics Overfitting Summary

Agenda

Regression basics

Linear regression

Metrics

Overfitting

Summary

Learning Outcomes

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- Explain overfitting and how regularization helps.

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- MAE/MSE/RMSE: regression error metrics
- R^2: coefficient of determination (variance explained)
- λ: regularization strength; $\|w\|_1, \|w\|_2$: L1/L2 penalties

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Examples:

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- The model is "linear in parameters".

Loss Function (MSE)

We learn parameters by minimizing a loss.

$$\text{MSE} = \frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2$$

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- Other losses exist (MAE, Huber), depending on needs.

Mini Worked Example (Compute Errors)

Suppose true values $y = [50, 60, 80]$ and predictions $\hat{y} = [55, 58, 70]$.

- Errors: $e = y - \hat{y} = [-5, 2, 10]$

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- MSE $= \frac{25+4+100}{3} = \frac{129}{3} = 43$
- RMSE $= \sqrt{43} \approx 6.56$ (same units as y)

Regression Evaluation Metrics

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- R^2: proportion of variance explained (can be misleading)

Underfitting vs Overfitting

- **Underfitting**: model too simple, high bias, poor train and test.

Remedies: more data, better features, regularization, cross-validation.

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- **Overfitting**: model too complex, low train error but high test error.

Remedies: more data, better features, regularization, cross-validation.

Regularization (Idea)

Add a penalty to discourage overly large weights:

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- Ridge (ℓ_2) shrinks weights.
- Lasso (ℓ_1) can push some weights to zero (feature selection effect).

Summary

- Regression predicts continuous values.
- Linear regression minimizes a loss such as MSE.
- Metrics: MAE, RMSE, R^2 (interpret carefully).
- Overfitting can be reduced with regularization and proper validation.

Sources: Mueller and Massaron, *Machine Learning for Dummies*, For Dummies, 2016.;James, Witten, Hastie, and Tibshirani, *An Introduction to Statistical Learning* (2e), selected chapters.;Scikit-learn documentation, accessed 2026-02-28.

Exit Questions

1. What is the difference between regression and classification?
2. Why is RMSE often easier to interpret than MSE?
3. What does R^2 measure and when can it be misleading?
4. Explain overfitting and list two mitigation strategies.